

JESS Thermodynamic Database v8.9

Chemical species in reactions with Niobium

24-Jan-23

Charge	CAS	Count	Molecular formula	Mol. mass
AlizarinRedS-3				
-3	---	52	C(14)H(5)O(7)S(1)	317.250
Ascorbic-2 L-Ascorbate ion; Ascorbate ion				
-2	63983-50-6	133	C(6)H(6)O(6)	174.110
B(s) Boron; Boron, rhombohedral; Boron, beta				
0	7440-42-8	111	B(1)	10.8100
Br2(l) Bromine, liquid				
0	---	285	Br(2)	159.808
C(s) Carbon; Graphite; Carbon, hexagonal				
0	7440-44-0	530	C(1)	12.0110
CDTA-4 trans-1,2-cyclohexylenedinitrilotetraacetate ion; trans-1,2-diaminocyclohexanetetraacetate ion; CDTA ion				
-4	22005-54-5	301	C(14)H(18)N(2)O(8)	342.306
Cl2(g) Chlorine gas				
0	7782-50-5	589	Cl(2)	70.9060
Cs+1 Caesium(I) ion; Cesium(I) ion				
1	18459-37-5	246	Cs(1)	132.910
Cs(s) Caesium; Cesium; Caesium, cubic				

0	7440-46-2	37	Cs (1)	132.910
CsNbUO6 (s) Cesium niobium uranium hexoxide				
0	---	2	Cs (1) Nb (1) O (6) U (1)	559.842
e-1 Electron				
-1	---	3399	E (1)	0.000000
EDTA-4 Ethylenediaminetetraacetate anion				
-4	---	1457	C (10) H (12) N (2) O (8)	288.214
F-1 Fluoride ion				
-1	16984-48-8	1090	F (1)	18.9984
F2 (g) Fluorine gas				
0	7782-41-4	259	F (2)	37.9968
H+1 Hydrogen ion; Proton				
1	12408-02-5	31679	H (1)	1.00794
H+1_34DHB-3				
-2	---	5	C (7) H (4) O (4)	152.106
H+1_F-1				
0	---	142	F (1) H (1)	20.0063
H+1_Na+1 (7)_Nb6O19-8_H2O (5)_ (s)				
0	---	1	H (11) Na (7) Nb (6) O (24)	1113.44
H+1_Nb6O19-8				
-7	---	6	H (1) Nb (6) O (19)	862.434
H+1_NbO3-1				

0	---	1	H(1)Nb(1)O(3)	141.912
H+1_Oxalic-2				
-1	---	38	C(2)H(1)O(4)	89.0275
H+1(10)_Nb12O36-12				
-2	---	1	H(10)Nb(12)O(36)	1700.93
H+1(2)_Asp-2 Aspartic acid; L-Aspartic acid; (S)-(+)-Aminosuccinic acid; Asparagic acid; 2-Aminobutanedioic acid; Aminobutanedioic acid				
0	---	8	C(4)H(7)N(1)O(4)	133.104
H+1(2)_Malic-2 Malic acid; Hydroxybutanedioic acid; L-Malic acid; L-Hydroxybutandioic acid; (S)-(-)-Hydroxysuccinic acid; (S)-(-)-Malic acid; 2-Hydroxysuccinic acid; Hydroxyethane-1,2-dicarboxylic acid				
0	97-67-6	18	C(4)H(6)O(5)	134.089
H+1(2)_Malonic-2				
0	---	4	C(3)H(4)O(4)	104.062
H+1(2)_Nb12O36-12				
-10	---	4	H(2)Nb(12)O(36)	1692.87
H+1(2)_Nb6O19-8				
-6	---	6	H(2)Nb(6)O(19)	863.442
H+1(2)_Oxalic-2				
0	144-62-7	33	C(2)H(2)O(4)	90.0355
H+1(2)_Succinic-2				
0	---	3	C(4)H(6)O(4)	118.089
H+1(2)_Tartaric-2 Tartaric acid; L-Tartaric acid; d-Tartaric acid (natural); dextroTartaric acid; 2,3-Dihydroxybutanedioic acid; Dihydroxysuccinic acid				
0	87-69-4	22	C(4)H(6)O(6)	150.088

H+1 (3) _34DHB-3 Protocatechuic acid; 3,4-Dihydroxybenzoic acid				
0	99-50-3	7	C (7) H (6) O (4)	154.122
H+1 (3) _EDTA-4				
-1	---	12	C (10) H (15) N (2) O (8)	291.238
H+1 (3) _Nb6O19-8				
-5	---	8	H (3) Nb (6) O (19)	864.450
H+1 (3) _SalHydroxam-2				
1	---	3	C (7) H (8) N (1) O (3)	154.145
H+1 (4) _EDTA-4 Edetic Acid; EDTA; Ethylenediaminetetraacetic acid; N,N'-1,2-Ethanediyldis[N-(carboxymethyl)glycine; Ethylenedinitrilotetraacetic acid; 1,2-Diamino-N,N,N',N'-ethane tetraacetic acid; Versene Acid; (1,2-Ethanediyldinitrilo)tetraacetic acid; H4edta; Sequestic acid				
0	60-00-4	13	C (10) H (16) N (2) O (8)	292.246
H+1 (7) _Nb12O36-12				
-5	---	1	H (7) Nb (12) O (36)	1697.91
H+1 (9) _Nb12O36-12				
-3	---	1	H (9) Nb (12) O (36)	1699.92
H2 (g) Hydrogen gas; Hydrogen				
0	1333-74-0	1564	H (2)	2.01588
H2O Water				
0	7732-18-5	5947	H (2) O (1)	18.0153
I2 (s) Iodine; Iodine, orthorhombic				
0	7553-56-2	244	I (2)	253.800
K+1 Potassium(I) ion				

1	24203-36-9	462	K (1)	39.0980
K (s) Potassium; Potassium, cubic				
0	7440-09-7	121	K (1)	39.0980
KNbO3 (s)				
0	---	1	K (1) Nb (1) O (3)	180.002
KNbUO6 (beta, s) Potassium niobium uranium hexoxide (beta)				
0	---	2	K (1) Nb (1) O (6) U (1)	466.030
KNbUO6 (s) Potassium niobium uranium hexoxide				
0	---	2	K (1) Nb (1) O (6) U (1)	466.030
Li (s) Lithium; Lithium, cubic				
0	7439-93-2	62	Li (1)	6.94100
LiNbO3 (s)				
0	---	1	Li (1) Nb (1) O (3)	147.845
N2 (g) Nitrogen gas; Nitrogen				
0	7727-37-9	566	N (2)	28.0134
Na+1 Sodium(I) ion				
1	17341-25-2	617	Na (1)	22.9900
Na (s) Sodium; Sodium, cubic				
0	7440-23-5	209	Na (1)	22.9900
NaNbO3 (s)				
0	---	1	Na (1) Nb (1) O (3)	163.894

Nb+3 Niobium(III) ion; Columbium(III) ion				
3	22541-83-9	10	Nb (1)	92.9062
Nb+3_Cl-1 (3) (s) Niobium trichloride				
0	---	1	Cl (3) Nb (1)	199.265
Nb+3_OH-1 (2)				
1	---	1	H (2) Nb (1) O (2)	126.921
Nb+3_OH-1 (4)				
-1	---	2	H (4) Nb (1) O (4)	160.936
Nb+4_Cl-1 (4) (s) Niobium tetrachloride				
0	---	1	Cl (4) Nb (1)	234.718
Nb+5				
5	---	81	Nb (1)	92.9062
Nb+5_Ascorbic-2				
3	---	1	C (6) H (6) Nb (1) O (6)	267.016
Nb+5_Br-1 (5) (s) Niobium pentabromide				
0	---	1	Br (5) Nb (1)	492.426
Nb+5_CDTA-4				
1	---	1	C (14) H (18) N (2) Nb (1) O (8)	435.212
Nb+5_Cl-1 (5) (g) Niobium pentachloride gas				
0	---	1	Cl (5) Nb (1)	270.171
Nb+5_Cl-1 (5) (s) Niobium pentachloride				
0	---	1	Cl (5) Nb (1)	270.171

Nb+5 _Cl-1 (5) _OH-1 (2)				
-2	---	1	Cl (5) H (2) Nb (1) O (2)	304.186
Nb+5 _Cs+1 _Cl-1 (6) _ (s)				
0	---	1	Cl (6) Cs (1) Nb (1)	438.534
Nb+5 _EDTA-4				
1	---	1	C (10) H (12) N (2) Nb (1) O (8)	381.120
Nb+5 _F-1				
4	---	1	F (1) Nb (1)	111.905
Nb+5 _F-1 _OH-1 (2)				
2	---	1	F (1) H (2) Nb (1) O (2)	145.919
Nb+5 _F-1 _OH-1 (4)				
0	---	1	F (1) H (4) Nb (1) O (4)	179.934
Nb+5 _F-1 _OH-1 (5)				
-1	---	1	F (1) H (5) Nb (1) O (5)	196.941
Nb+5 _F-1 (2) _OH-1 (2)				
1	---	2	F (2) H (2) Nb (1) O (2)	164.918
Nb+5 _F-1 (2) _OH-1 (3)				
0	---	3	F (2) H (3) Nb (1) O (3)	181.925
Nb+5 _F-1 (2) _OH-1 (4)				
-1	---	1	F (2) H (4) Nb (1) O (4)	198.932
Nb+5 _F-1 (3) _OH-1 (2)				
0	---	3	F (3) H (2) Nb (1) O (2)	183.916
Nb+5 _F-1 (4) _OH-1 (2)				
-1	---	2	F (4) H (2) Nb (1) O (2)	202.915

Nb+5_F-1 (5)				
0	---	1	F (5) Nb (1)	187.898
Nb+5_F-1 (5)_(s) Niobium pentafluoride				
0	---	1	F (5) Nb (1)	187.898
Nb+5_F-1 (5)_OH-1 (2)				
-2	---	2	F (5) H (2) Nb (1) O (2)	221.913
Nb+5_F-1 (6)_OH-1 (2)				
-3	---	2	F (6) H (2) Nb (1) O (2)	240.911
Nb+5_F-1 (7)				
-2	---	3	F (7) Nb (1)	225.895
Nb+5_F-1 (8)				
-3	---	2	F (8) Nb (1)	244.893
Nb+5_F-1 (9)				
-4	---	1	F (9) Nb (1)	263.892
Nb+5_H+1_OH-1 (4)_Malic-2				
0	---	1	C (4) H (9) Nb (1) O (9)	294.016
Nb+5_H+1_OH-1 (4)_Malonic-2				
0	---	1	C (3) H (7) Nb (1) O (8)	263.990
Nb+5_H+1_OH-1 (4)_Oxalic-2				
0	---	1	C (2) H (5) Nb (1) O (8)	249.963
Nb+5_H+1_OH-1 (4)_Succinic-2				
0	---	1	C (4) H (9) Nb (1) O (8)	278.017
Nb+5_H+1_OH-1 (4)_Tartaric-2				
0	---	1	C (4) H (9) Nb (1) O (10)	310.016

Nb+5_H+1(2)_OH-1(4)_34DHB-3				
0	---	2	C(7)H(9)Nb(1)O(8)	314.050
Nb+5_H+1(2)_OH-1(4)_Asp-2				
1	---	1	C(4)H(11)N(1)Nb(1)O(8)	294.039
Nb+5_H+1(3)_OH-1(4)_EDTA-4				
0	---	1	C(10)H(19)N(2)Nb(1)O(12)	452.173
Nb+5_H+1(4)_OH-1(4)_34DHB-3(2)				
-1	---	1	C(14)H(14)Nb(1)O(12)	467.164
Nb+5_H+1(6)_Cl-1(5)_OH-1(2)_SalHydroxam-2(2)				
0	---	1	C(14)H(18)Cl(5)N(2)Nb(1)O(8)	612.477
Nb+5_H+1(7)_OH-1(4)_34DHB-3(3)				
-1	---	1	C(21)H(20)Nb(1)O(16)	621.286
Nb+5_H2O2				
5	---	4	H(2)Nb(1)O(2)	126.921
Nb+5_H2O2_F-1				
4	---	1	F(1)H(2)Nb(1)O(2)	145.919
Nb+5_H2O2_F-1(2)				
3	---	1	F(2)H(2)Nb(1)O(2)	164.918
Nb+5_H2O2_F-1(3)				
2	---	1	F(3)H(2)Nb(1)O(2)	183.916
Nb+5_H2O2_F-1(4)				
1	---	1	F(4)H(2)Nb(1)O(2)	202.915
Nb+5_I-1(5)_(s) Niobium pentaiodide				
0	---	1	I(5)Nb(1)	727.406

Nb+5_K+1_Cl-1(6)_(s)				
0	---	1	Cl(6)K(1)Nb(1)	344.722
Nb+5_Na+1_Cl-1(6)_(s)				
0	---	1	Cl(6)Na(1)Nb(1)	328.614
Nb+5_OH-1				
4	---	2	H(1)Nb(1)O(1)	109.914
Nb+5_OH-1(2)				
3	---	8	H(2)Nb(1)O(2)	126.921
Nb+5_OH-1(2)_EDTA-4				
-1	---	2	C(10)H(14)N(2)Nb(1)O(10)	415.135
Nb+5_OH-1(2)_Oxalic-2(2)				
-1	---	5	C(4)H(2)Nb(1)O(10)	302.960
Nb+5_OH-1(2)_Oxalic-2(3)				
-3	---	3	C(6)H(2)Nb(1)O(14)	390.980
Nb+5_OH-1(2)_Salicylic-2(2)				
-1	---	1	C(14)H(10)Nb(1)O(8)	399.135
Nb+5_OH-1(2)_Tartaric-2				
1	---	2	C(4)H(6)Nb(1)O(8)	274.993
Nb+5_OH-1(2)_Tartaric-2(2)				
-1	---	2	C(8)H(10)Nb(1)O(14)	423.065
Nb+5_OH-1(3)				
2	---	2	H(3)Nb(1)O(3)	143.928
Nb+5_OH-1(4)				
1	---	22	H(4)Nb(1)O(4)	160.936

Nb+5_OH-1(4)_Oxalic-2				
-1	---	1	C(2)H(4)Nb(1)O(8)	248.955
Nb+5_OH-1(5)				
0	---	12	H(5)Nb(1)O(5)	177.943
Nb+5_OH-1(5)_(am.,s)				
0	---	4	H(5)Nb(1)O(5)	177.943
Nb+5_OH-1(6)				
-1	---	4	H(6)Nb(1)O(6)	194.950
Nb+5_OH-1(7)				
-2	---	1	H(7)Nb(1)O(7)	211.958
Nb+5_OH-1(8)				
-3	---	2	H(8)Nb(1)O(8)	228.965
Nb+5_Rb+1_Cl-1(6)_(s)				
0	---	1	Cl(6)Nb(1)Rb(1)	391.092
Nb(s) Niobium				
0	7440-03-1	49	Nb(1)	92.9062
Nb2C(s) Diniobium carbide				
0	---	1	C(1)Nb(2)	197.823
Nb2N(s) Diniobium nitride				
0	---	1	N(1)Nb(2)	199.819
Nb2O2(s)				
0	---	1	Nb(2)O(2)	217.811
Nb2O3(s) Diniobium pentoxide				

0	---	1	Nb (2) O (3)	233.811
Nb₂O₅ (s) Diniobium pentaoxide				
0	---	10	Nb (2) O (5)	265.809
Nb₄O₁₆-12				
-12	---	2	Nb (4) O (16)	627.615
Nb₆O₁₉-8 polyoxoniobate				
-8	---	19	Nb (6) O (19)	861.426
NbB₂ (s) Niobium diboride				
0	---	1	B (2) Nb (1)	114.526
NbC (s) Niobium carbide				
0	---	1	C (1) Nb (1)	104.917
NbCl_{2.33} (s) Niobium 2.33-chloride				
0	---	1	Cl (2.33) Nb (1)	175.512
NbCl_{2.67} (s) Niobium 2.67-chloride				
0	---	1	Cl (2.67) Nb (1)	187.566
NbCl₂ (s) Niobium dichloride				
0	---	1	Cl (2) Nb (1)	163.812
NbCl_{3.13} (s) Niobium 3.13-chloride				
0	---	1	Cl (3.13) Nb (1)	203.874
NbN (s) Niobium nitride				
0	---	1	N (1) Nb (1)	106.913

NbO+2				
2	---	1	Nb (1) O (1)	108.906
NbO+3				
3	---	4	Nb (1) O (1)	108.906
NbO+3_AlizarinRedS-3 (2)				
-3	---	1	C (28) H (10) Nb (1) O (15) S (2)	743.405
NbO+3_PAR-2_Tartaric-2				
-1	---	1	C (15) H (11) N (3) Nb (1) O (9)	470.173
NbO+3_SO4-2 (2)				
-1	---	1	Nb (1) O (9) S (2)	301.021
NbO+3_Tartaric-2				
1	---	1	C (4) H (4) Nb (1) O (7)	256.978
NbO (s) Niobium monoxide				
0	---	4	Nb (1) O (1)	108.906
NbO2+1				
1	---	1	Nb (1) O (2)	124.905
NbO2 (s) Niobium dioxide				
0	---	6	Nb (1) O (2)	124.905
NbO2Cl (s) Niobium chloride dioxide				
0	---	1	Cl (1) Nb (1) O (2)	160.358
NbO3-1				
-1	---	6	Nb (1) O (3)	140.904
NbO6-7				

-7	---	1	Nb(1)O(6)	188.903
NbOBr3(s)				
0	---	1	Br(3)Nb(1)O(1)	348.618
NbOCl2(s) Niobium dichloride oxide				
0	---	1	Cl(2)Nb(1)O(1)	179.812
NbOCl3(s) Niobium trichloride oxide				
0	---	1	Cl(3)Nb(1)O(1)	215.265
NbOF3 Niobium oxyfluoride				
0	---	4	F(3)Nb(1)O(1)	165.901
O2(g) Oxygen gas; Oxygen				
0	7782-44-7	2559	O(2)	31.9988
OH-1 Hydroxide ion				
-1	14280-30-9	6558	H(1)O(1)	17.0073
OH-1(4)_Nb4O12-4				
-8	---	2	H(4)Nb(4)O(16)	631.647
Oxalic-2 Oxalate ion				
-2	338-70-5	1241	C(2)O(4)	88.0196
PAR-2 Pyridylazoresorcinolate ion				
-2	---	110	C(11)H(7)N(3)O(2)	213.196
Rb+1 Rubidium(I) ion				
1	22537-38-8	125	Rb(1)	85.4680

Rb (s) Rubidium; Rubidium, cubic				
0	7440-17-7	29	Rb(1)	85.4680
RbNbUO6 (s) Rubidium niobium uranium hexoxide				
0	---	2	Nb(1)O(6)Rb(1)U(1)	512.400
Salicylic-2 Salicylate ion				
-2	16887-56-2	558	C(7)H(4)O(3)	136.107
SO4-2 Sulfate ion; Sulphate ion				
-2	14808-79-8	1270	O(4)S(1)	96.0576
Tartaric-2 Tartrate ion				
-2	---	393	C(4)H(4)O(6)	148.072
U (s) Uranium; Uranium, orthorhombic; Uranium, alpha				
0	7440-61-1	263	U(1)	238.029
UO2+2 Uranyl ion; U(VI) cation; Uranate				
2	16637-16-4	1268	O(2)U(1)	270.028